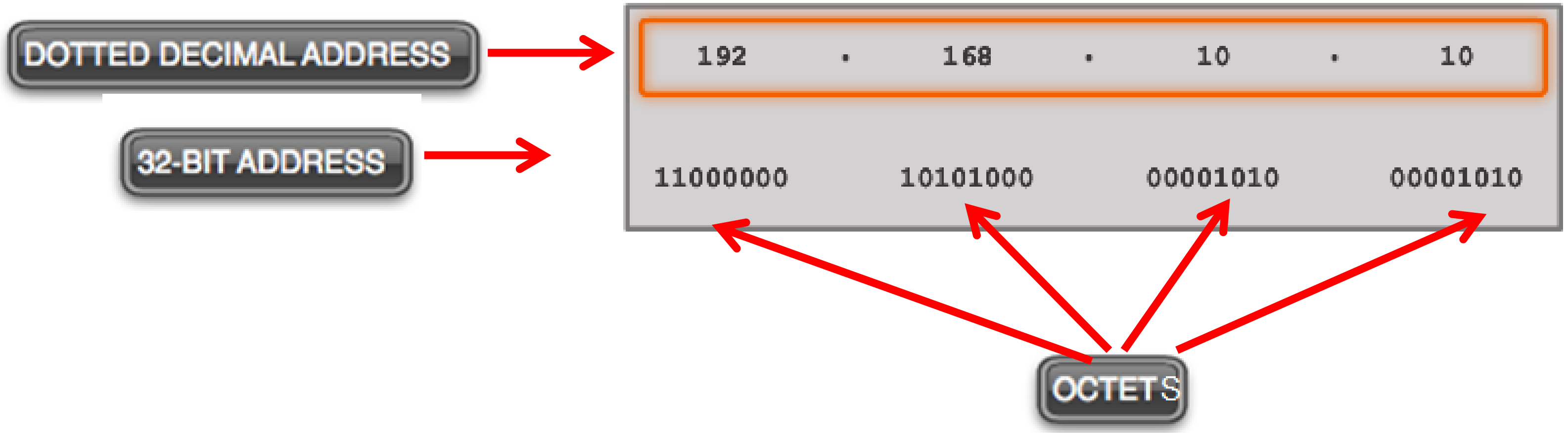
Network Fundamentals

IPv4 addressing and subnetting

IPv4 Address Structure

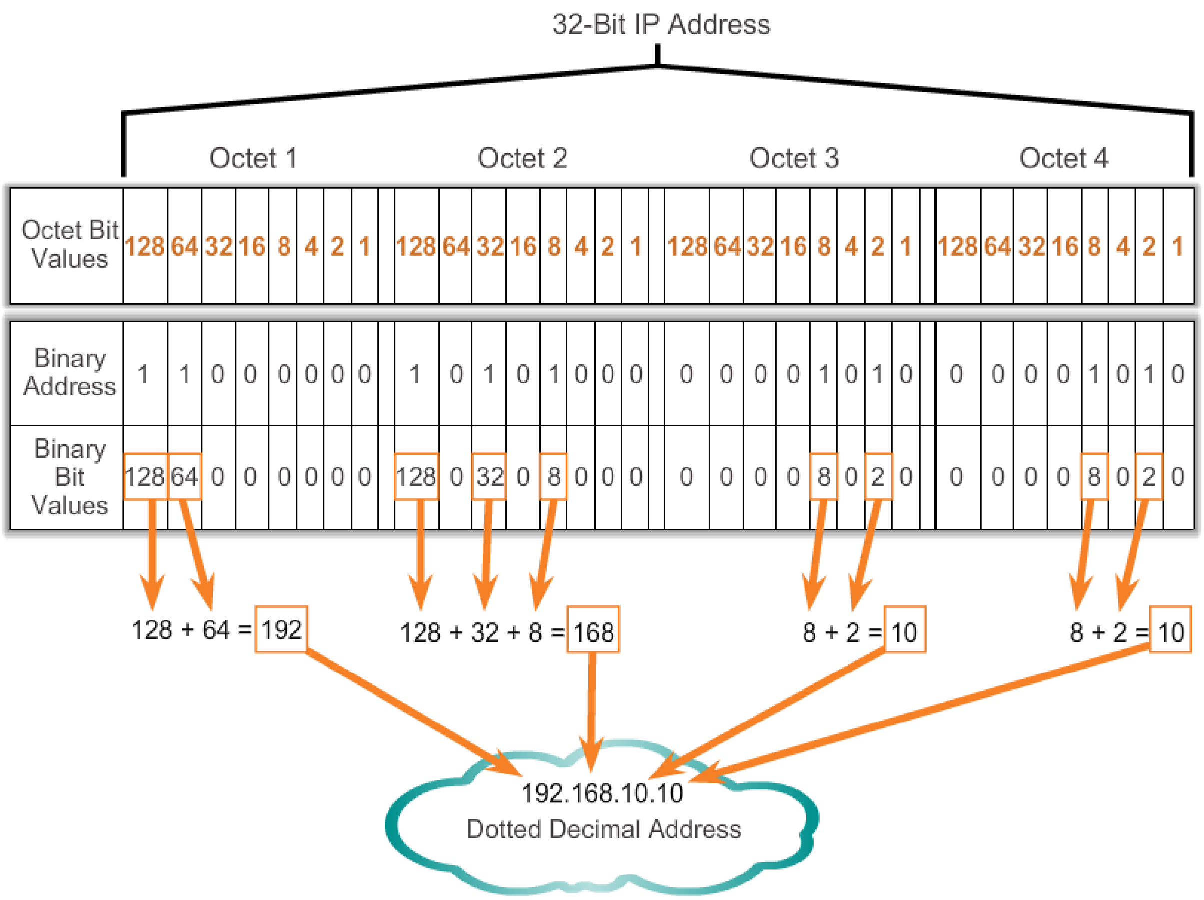


192.168.10.10 is an IP address that is assigned to a computer.

Radix	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Octet Bit Values	128	64	32	16	8	4	2	1
Binary Address	1	1	0	0	0	0	0	0
Binary Bit Values	128	64	0	0	0	0	0	0

Add the binary bit values. 128 + 64 = 192

Converting a Binary Address to Decimal



Converting a Binary Address to Decimal

Practice

Enter decimal answer here

Decimal value	228							
Radix	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Position	128	64	32	16	8	4	2	1
Bit	**1**	**1**	**1**	**0**	**0**	**1**	**0**	**0**

Binary number

Converting a Binary Address to Decimal

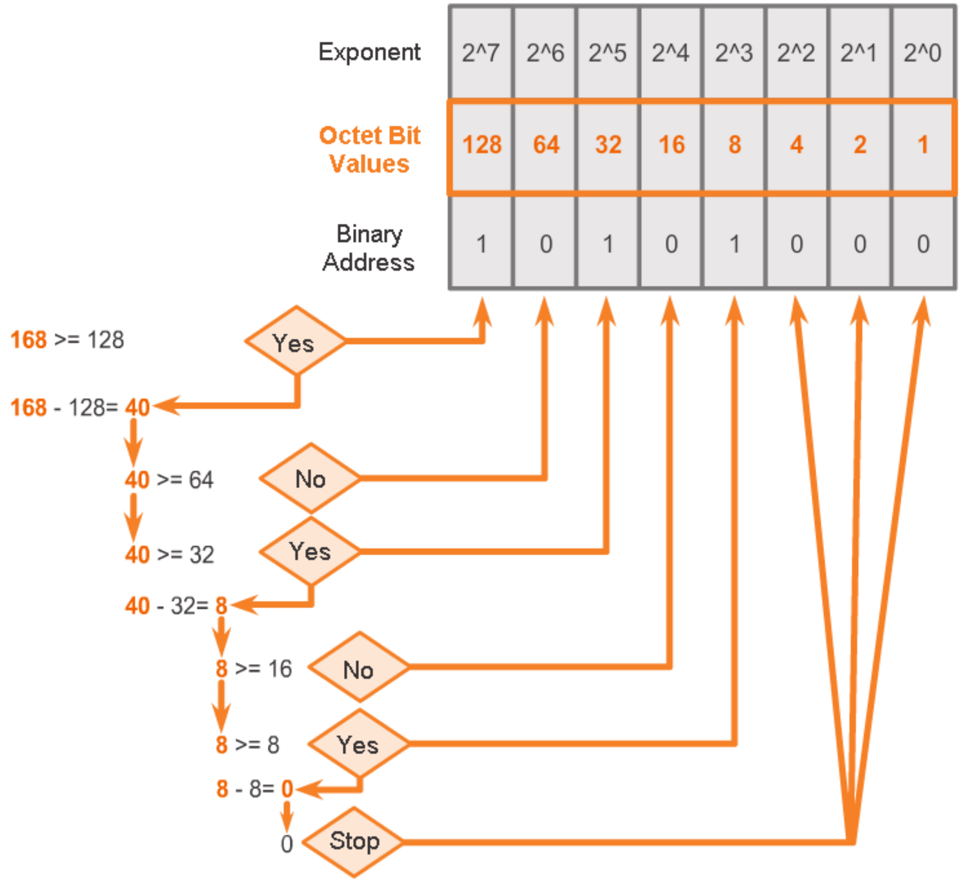
Practice

Enter decimal answer here

Decimal value	55							
Radix	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Position	128	64	32	16	8	4	2	1
Bit	**0**	**0**	**1**	**1**	**0**	**1**	**1**	**1**

Binary number

Converting from Decimal to Binary



Converting from Decimal to Binary

Convert Decimal to Binary

192.168.10.10

11000000

	128	64	32	16	8	4	2	1
192 > 128, place a 1 in the 128 position	1							
-128 subtract 128								
64 = 64, place a 1 in the 64 position	1	1						
-64 subtract 64								
0 place a 0 in all remaining positions.								
All done. Result	1	1	0	0	0	0	0	0

Converting from Decimal to Binary

Convert Decimal to Binary

192.**168**.10.10

11000000 **10101000**

	128	64	32	16	8	4	2	1
168 > 128, place a 1 in the 128 position	1							
-128 subtract 128								
40 < 64, place a 0 in the 64 position	1	0						
do not subtract								
40 > 32, place a 1 in the 32 position	1	0	1					
-32 subtract 32								
8 < 16, place a 0 in the 16 position	1	0	1	0				
do not subtract								
8 = 8 , place a 1 in the 8 position	1	0	1	0	1			
subtract 8								
0 place a 0 in all remaining positions								
All done. Result	**1**	**0**	**1**	**0**	**1**	**0**	**0**	**0**

Converting from Decimal to Binary

Convert Decimal to Binary

192.168.10.10

11000000 10101000 00001010

	128	64	32	16	8	4	2	1
10 < 128, place a 0 in the 128 position do not subtract	0	0	0	0	1	0	1	0
10 < 64, place a 0 in the 64 position do not subtract	0	0	0	0	1	0	1	0
10 < 32, place a 0 in the 32 position do not subtract	0	0	0	0	1	0	1	0
10 < 16, place a 0 in the 16 position do not subtract	0	0	0	0	1	0	1	0
10 > 8, place a 1 in the 8 position subtract 8	0	0	0	0	1	0	1	0
2 < 4, place a 0 in the 4 position do not subtract	0	0	0	0	1	0	1	0
2 = 2, place a 1 in the 2 position -2 subtract 2	0	0	0	0	1	0	1	0
0 place a 0 in all remaining positions All done. Result	0	0	0	0	1	0	1	0

Converting from Decimal to Binary Conversions

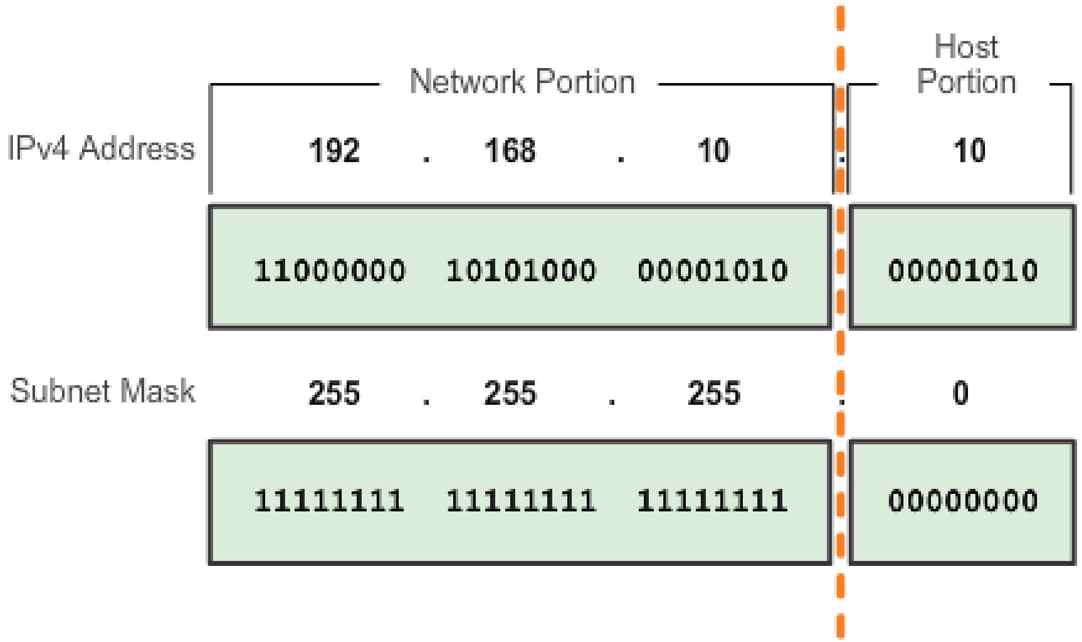
Practice:

- https://www.networkacademy.io/ccna/ip-subnetting/converting-ip-addresses-into-binary
- https://ccnax.com/free-quiz/

Decimal value	230							
Radix	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Position	128	64	32	16	8	4	2	1
Bit	1	1	1	0	0	1	1	0

Network Portion and Host Portion of an IPv4 Address

- To define the network and host portions of an address, a devices use a separate 32-bit pattern called a **subnet mask**
- The subnet mask does not actually contain the network or host portion of an IPv4 address, it just says where to look for these portions in a given IPv4 address



Network Portion and Host Portion of an IPv4 Address

The **bits within the network portion of the address**
→ **identical** for all devices in the same network
The **bits within the host portion of the address**
→ **unique** to identify a specific host within a network

If two hosts have the **same bit-pattern** in the network portion
→ those two hosts are in the same network and can communicate !

How do hosts know which portion the network portion is and which the host?
→ Job of the **subnet mask**
→ when an IP host is configured, a subnet mask assigned along
→ like the IP address, the subnet mask is 32 bits long
→ the subnet mask signifies which part of the IP address is network and which part is host

Network Portion and Host Portion of an IPv4 Address

The subnet mask is compared to the IP address from left to right, bit for bit

→ the **1s** represent the **network** portion;
the **0s** represent the **host** portion

Creating subnet mask

→ placing a binary 1 in each bit position of the network portion

→ placing a binary 0 in each bit position of the host portion

Example:

255.255.255.252

1111 1111 1111 1111 1111 1111 1111 1100

1111 1111 1111 1111 1111 1111 1111 1100

Network host

Valid Subnet Masks

Subnet Mask	Network bits	# of Host per Subnet
255.255.255.252	/30	2
255.255.255.248	/29	6
255.255.255.240	/28	14
255.255.255.224	/27	30
255.255.255.192	/26	62
255.255.255.128	/25	126
255.255.255.0	/24	254
255.255.254.0	/23	510
255.255.252.0	/22	1,022
255.255.248.0	/21	2,046
255.255.240.0	/20	4,094
255.255.224.0	/19	8,190
255.255.192.0	/18	16,382
255.255.128.0	/17	32,766
255.255.0.0	/16	65,534
255.254.0.0	/15	131,070
255.252.0.0	/14	262,142
255.248.0.0	/13	524,286
255.240.0.0	/12	1,048,574
255.224.0.0	/11	2,097,150
255.192.0.0	/10	4,194,302
255.128.0.0	/9	8,288,606
255.0.0.0	/8	16,777,216

Subnet Value	Bit Value							
	128	64	32	16	8	4	2	1
255	1	1	1	1	1	1	1	1
254	1	1	1	1	1	1	1	0
252	1	1	1	1	1	1	0	0
248	1	1	1	1	1	0	0	0
240	1	1	1	1	0	0	0	0
224	1	1	1	0	0	0	0	0
192	1	1	0	0	0	0	0	0
128	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

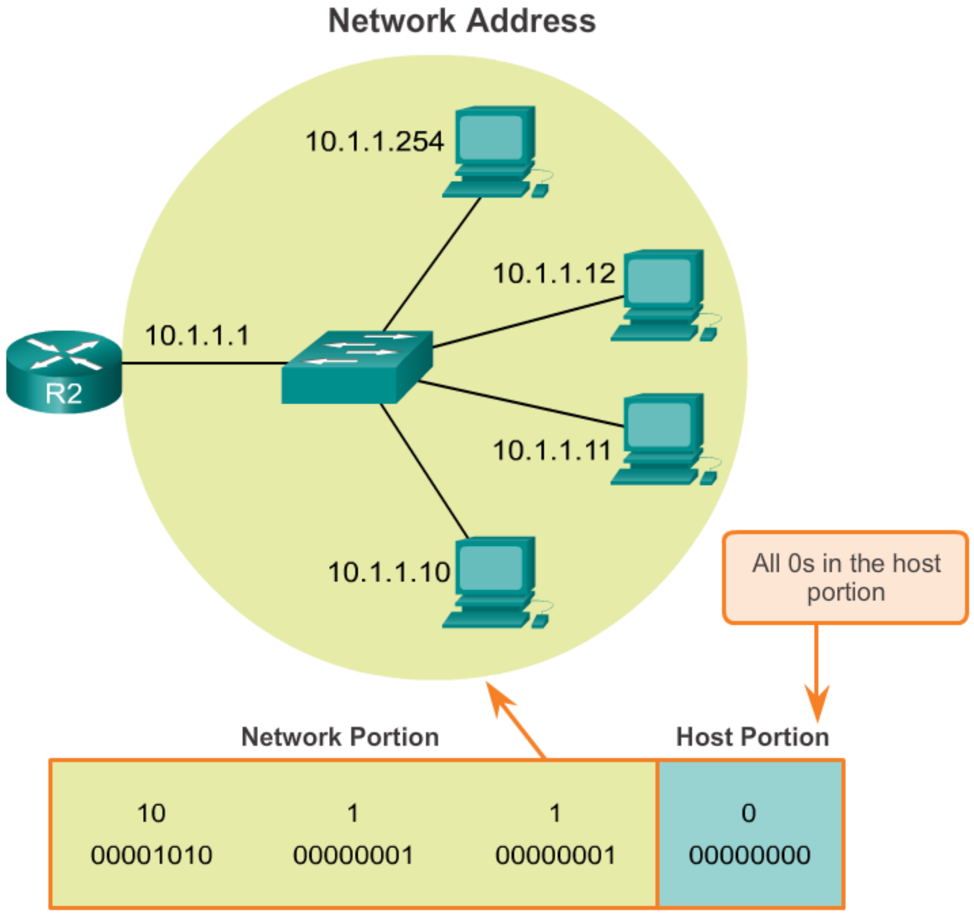
the Prefix Length

Prefix length
→ another way of expressing the subnet mask
→ = the number of bits set to 1 in the subnet mask
→ “slash notation”, a “/” followed by the number of bits set to 1

Example:
Ip address: 192.168.1.10/24
the subnet mask: 255.255.255.0
- 24 bits set to 1 in the binary version of the subnet mask
- prefix length is 24 bits or /24

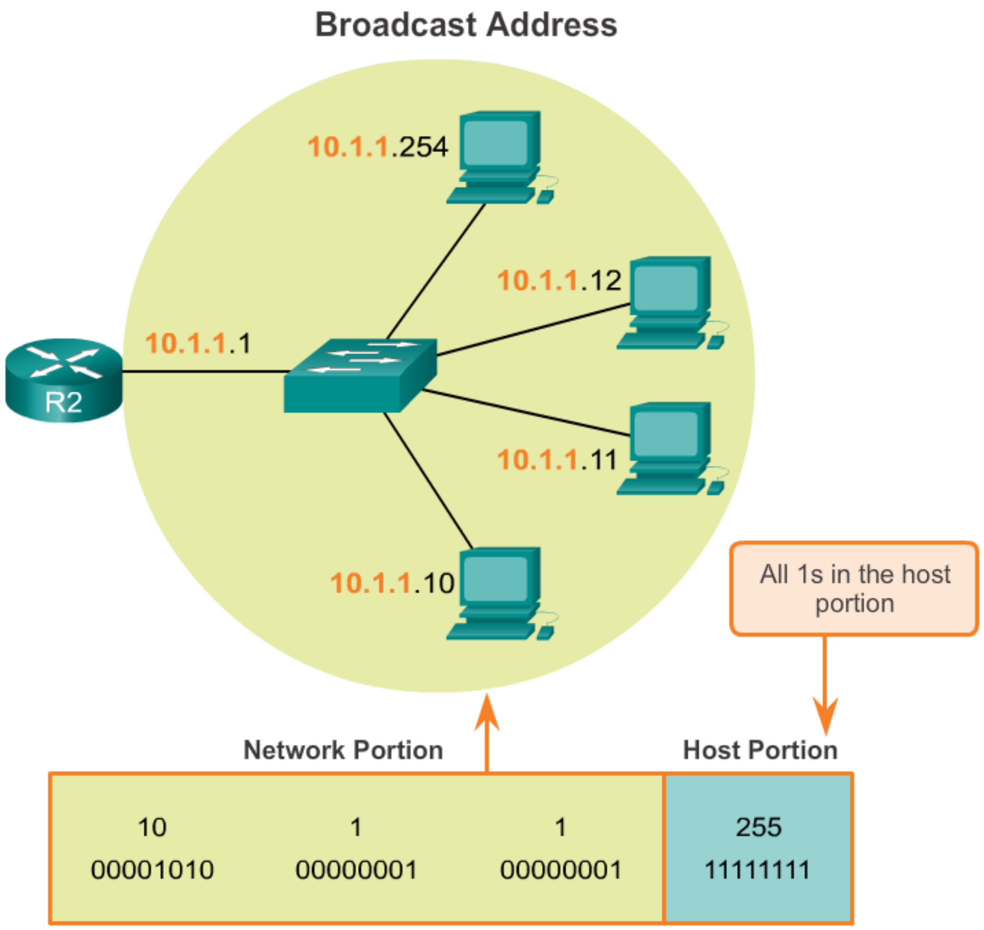
IPv4 Network, Host, and Broadcast Address

10.1.1.0/24 network



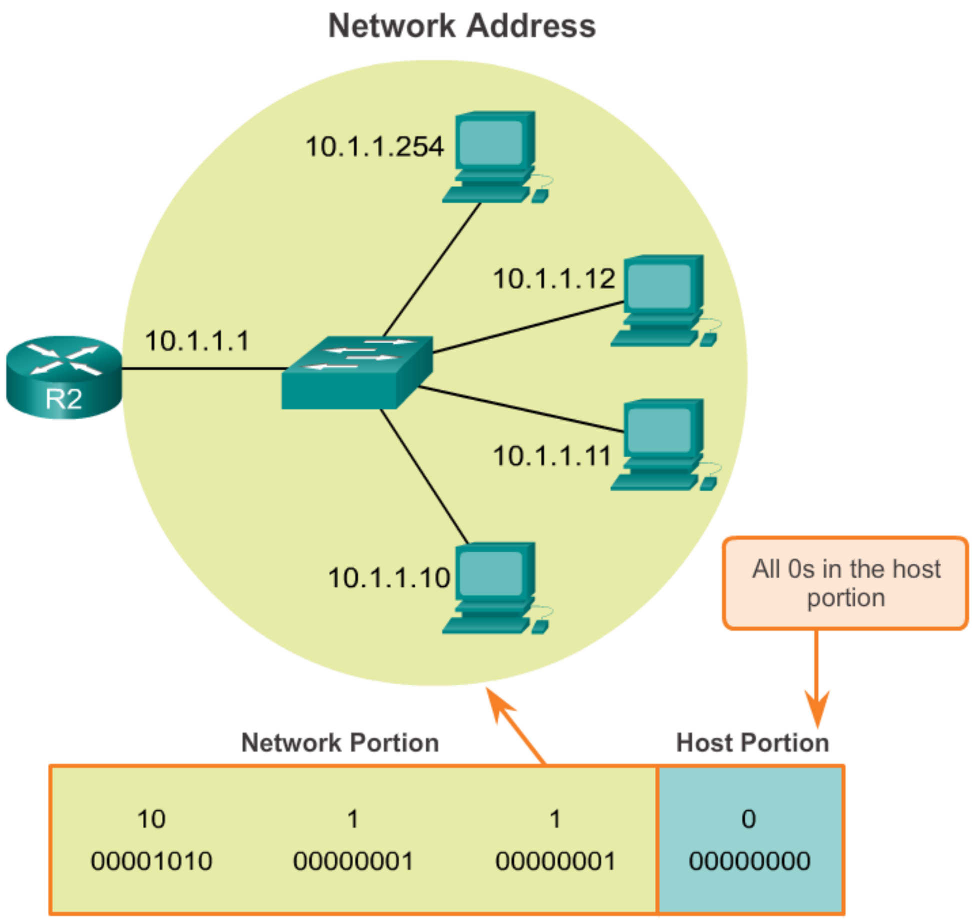
IPv4 Network, Host, and Broadcast Address

10.1.1.0/24 network



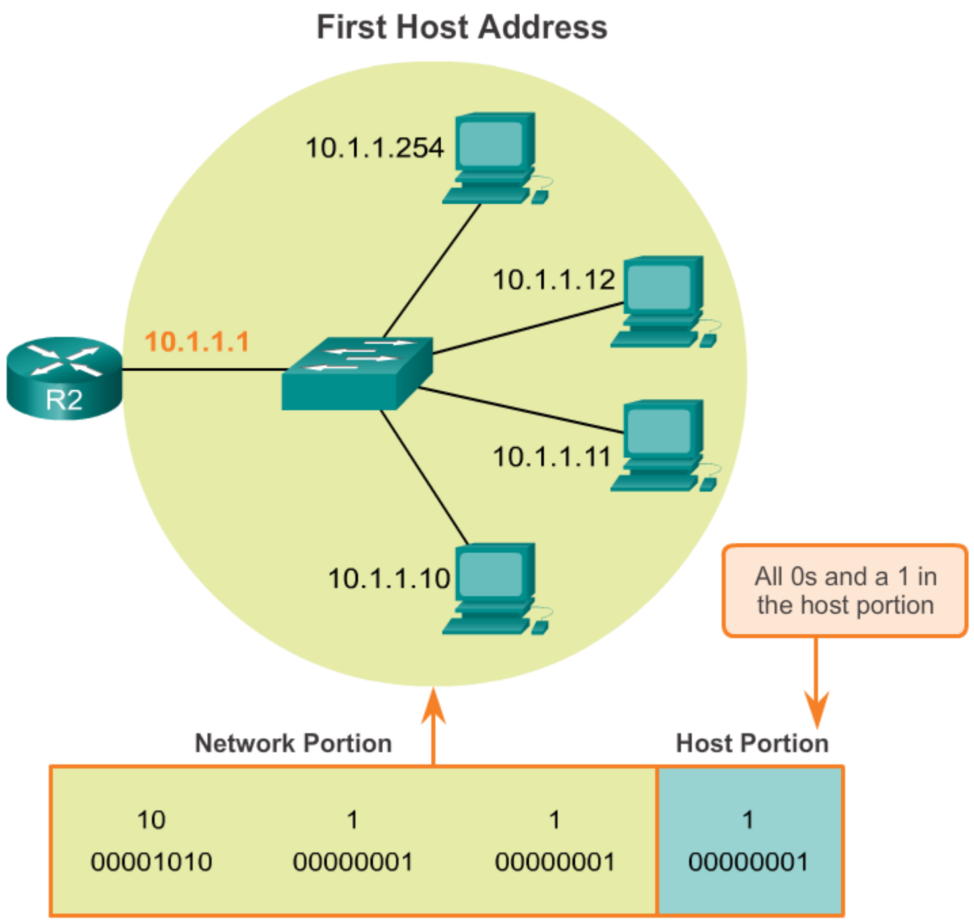
IPv4 Network, Host, and Broadcast Address

10.1.1.0/24 network



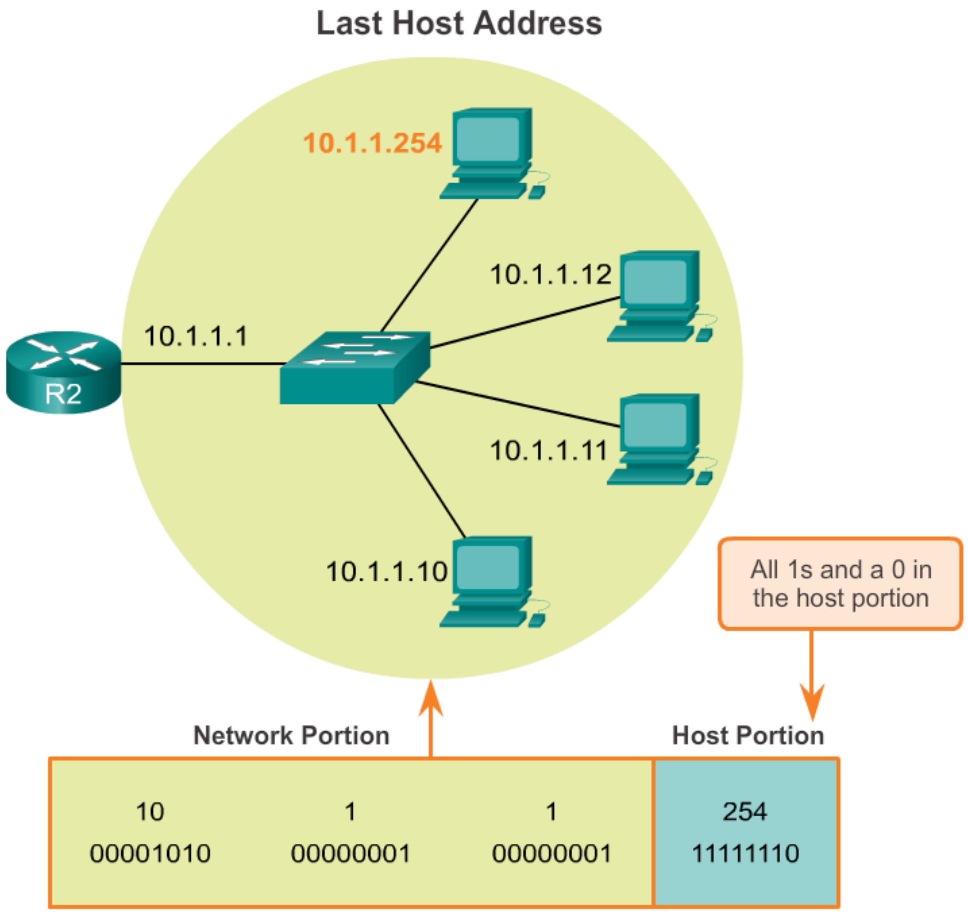
IPv4 Network, Host, and Broadcast Address

10.1.1.0/24 network



IPv4 Network, Host, and Broadcast Address

10.1.1.0/24 network



the Prefix Length

	Dotted Decimal	Significant bits shown in binary
Network Address	**10.1.1.0/24**	**10.1.1.00000000**
First Host Address	10.1.1.1	10.1.1.00000001
Last Host Address	10.1.1.254	10.1.1.11111110
Broadcast Address	10.1.1.255	10.1.1.11111111
Number of hosts: 2^8 – 2 = 254 hosts		

Network Address	**10.1.1.0/25**	**10.1.1.00000000**
First Host Address	10.1.1.1	10.1.1.00000001
Last Host Address	10.1.1.126	10.1.1.01111110
Broadcast Address	10.1.1.127	10.1.1.01111111
Number of hosts: 2^7 – 2 = 126 hosts		

Network Address	**10.1.1.0/26**	**10.1.1.00000000**
First Host Address	10.1.1.1	10.1.1.00000001
Last Host Address	10.1.1.62	10.1.1.00111110
Broadcast Address	10.1.1.63	10.1.1.00111111
Number of hosts: 2^6 – 2 = 62 hosts		

	Dotted Decimal	Significant bits shown in binary
Network Address	**10.1.1.0/27**	**10.1.1.00000000**
First Host Address	10.1.1.1	10.1.1.00000001
Last Host Address	10.1.1.30	10.1.1.00011110
Broadcast Address	10.1.1.31	10.1.1.00011111
Number of hosts: 2^5 – 2 = 30 hosts		

Network Address	**10.1.1.0/28**	**10.1.1.00000000**
First Host Address	10.1.1.1	10.1.1.00000001
Last Host Address	10.1.1.14	10.1.1.00001110
Broadcast Address	10.1.1.15	10.1.1.00001111
Number of hosts: 2^4 – 2 = 14 hosts		

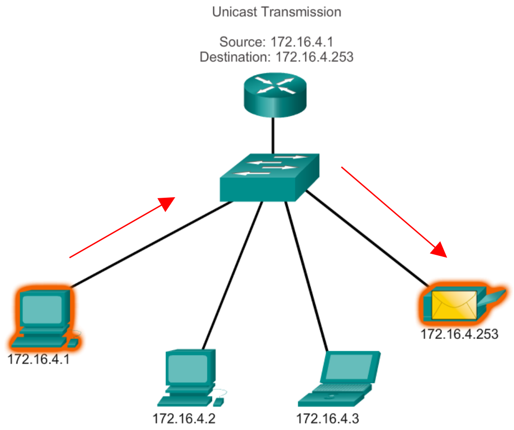
IPv4 Unicast, Broadcast, and Multicast

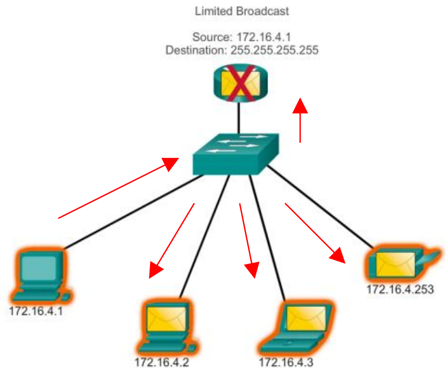
Unicast - the process of sending a packet from one host to an individual host.

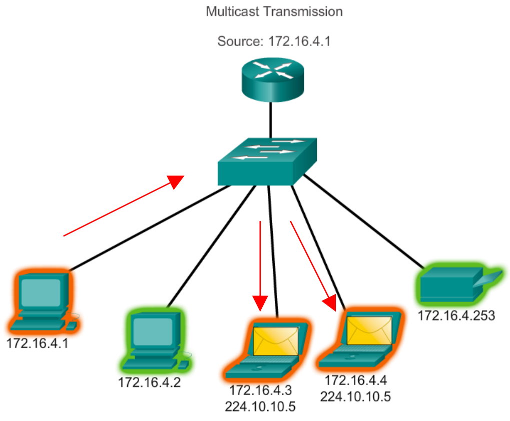
Broadcast - the process of sending a packet from one host to all hosts in the network

Multicast - the process of sending a packet from one host to a selected group of hosts, possibly in different networks

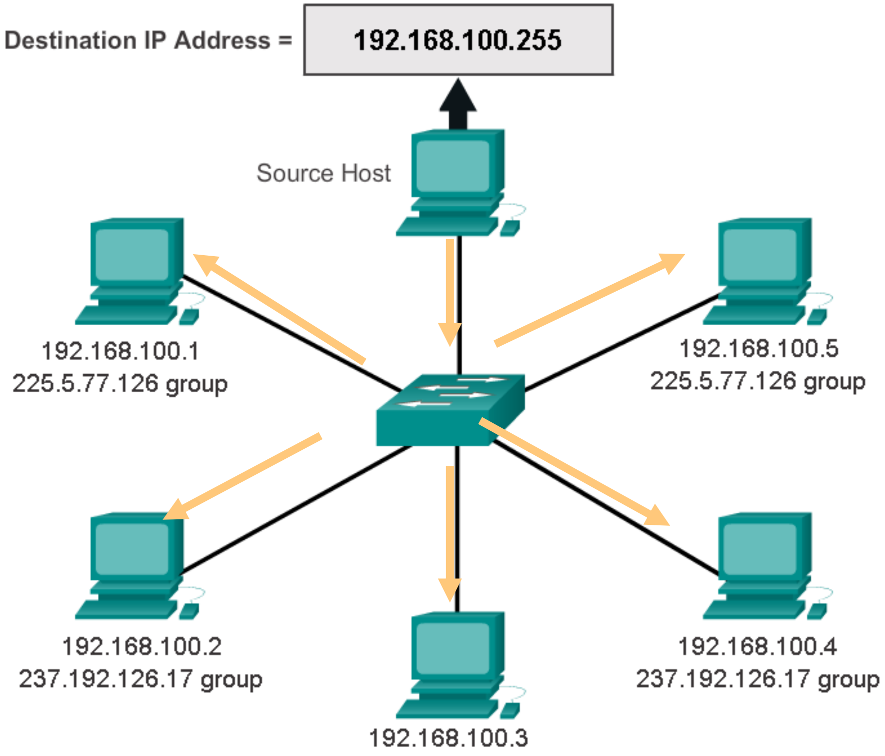
- Reserved for addressing multicast groups - 224.0.0.0 to 239.255.255.255.

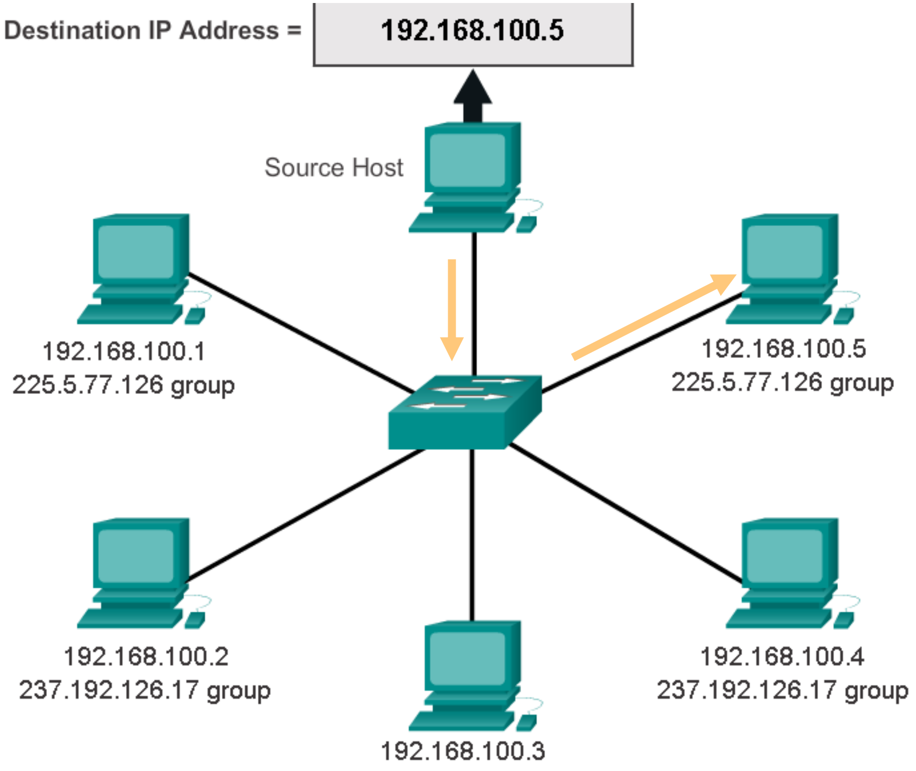




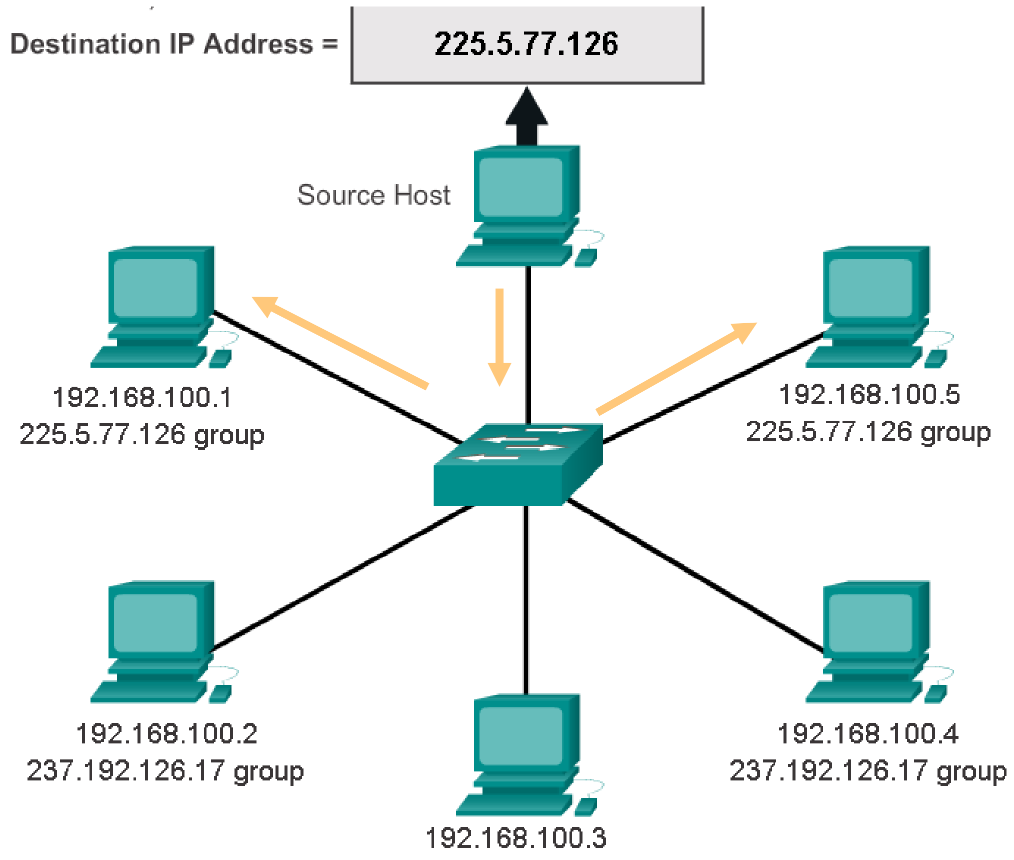


IPv4 Unicast, Broadcast, and Multicast





IPv4 Unicast, Broadcast, and Multicast

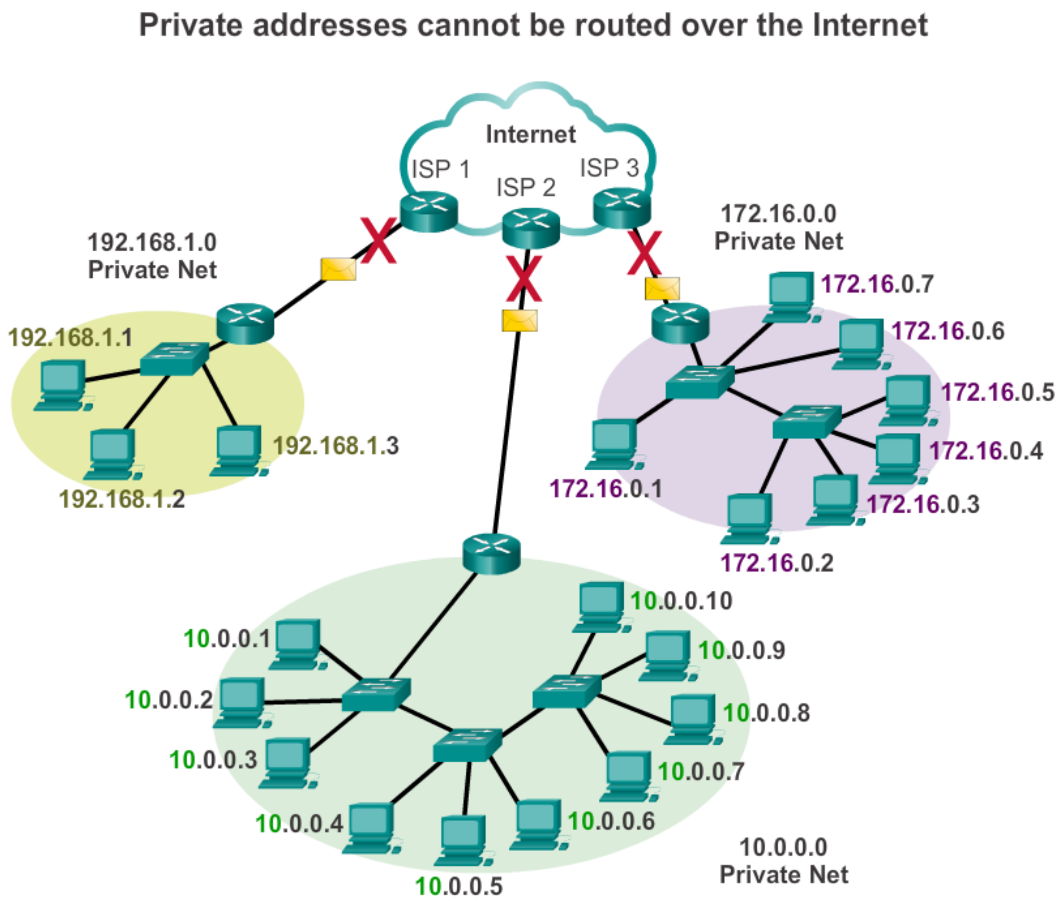


Public and Private IPv4 Addresses

Private address blocks are:
Hosts on a private network can use private addresses

- 10.0.0.0/8 10.0.0.0 → 10.255.255.255
- 172.16.0.0/12 172.16.0.0 → 172.31.255.255
- 192.168.0.0/16 192.168.0.0 → 192.168.255.255

Public and Private IPv4 Addresses



Special Use IPv4 Addresses

- **Network and Broadcast addresses** - within each network the **first** and **last** addresses cannot be assigned to hosts
- **Loopback address -** 127.0.0.1 a special address that hosts use to direct traffic to themselves (addresses 127.0.0.0 to 127.255.255.255 are reserved)
- **Link-Local address -** 169.254.0.0 to 169.254.255.255 (169.254.0.0/16) addresses can be automatically assigned to the local host (with no DHCP server) (TTL = 1)
- **TEST-NET addresses** - 192.0.2.0 to 192.0.2.255 (192.0.2.0/24) set aside for teaching and learning purposes, used in documentation and network examples
- **Experimental addresses -** 240.0.0.0 to 255.255.255.254 are listed as reserved

Assigning IP addresses to subnets

Start with IP address pool

- 192.168.0.0/16 → 65 536 addresses

Assinging IP addresses to subnets

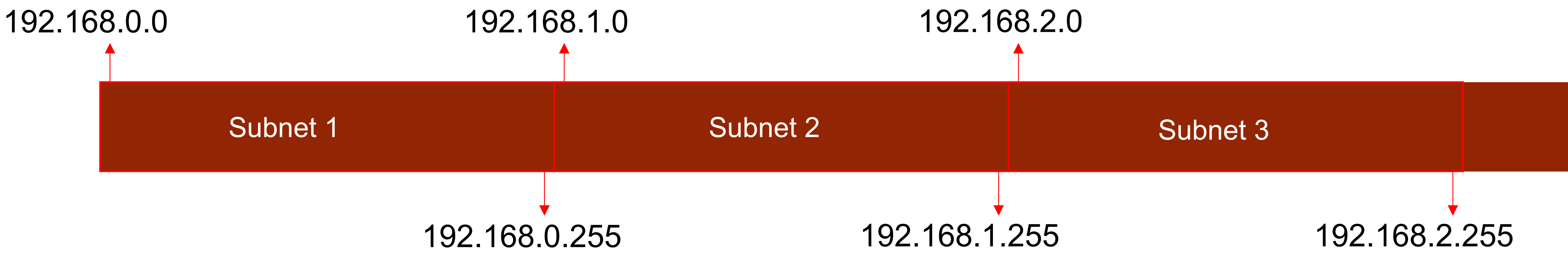
You can for example divide this address pool into 256 networks by changing the subnet mask

192.168.0.0/16

Assinging IP addresses to subnets

192.168. 0 . 0 /16 → /24

256 subnets 256 addresses per subnet



Assinging IP addresses to subnets

Subnet 1

192.168.0.0	Network address
192.168.0.1	First host address
192.168.0.254	Last host address
192.168.0.255	Subnet broadcast address

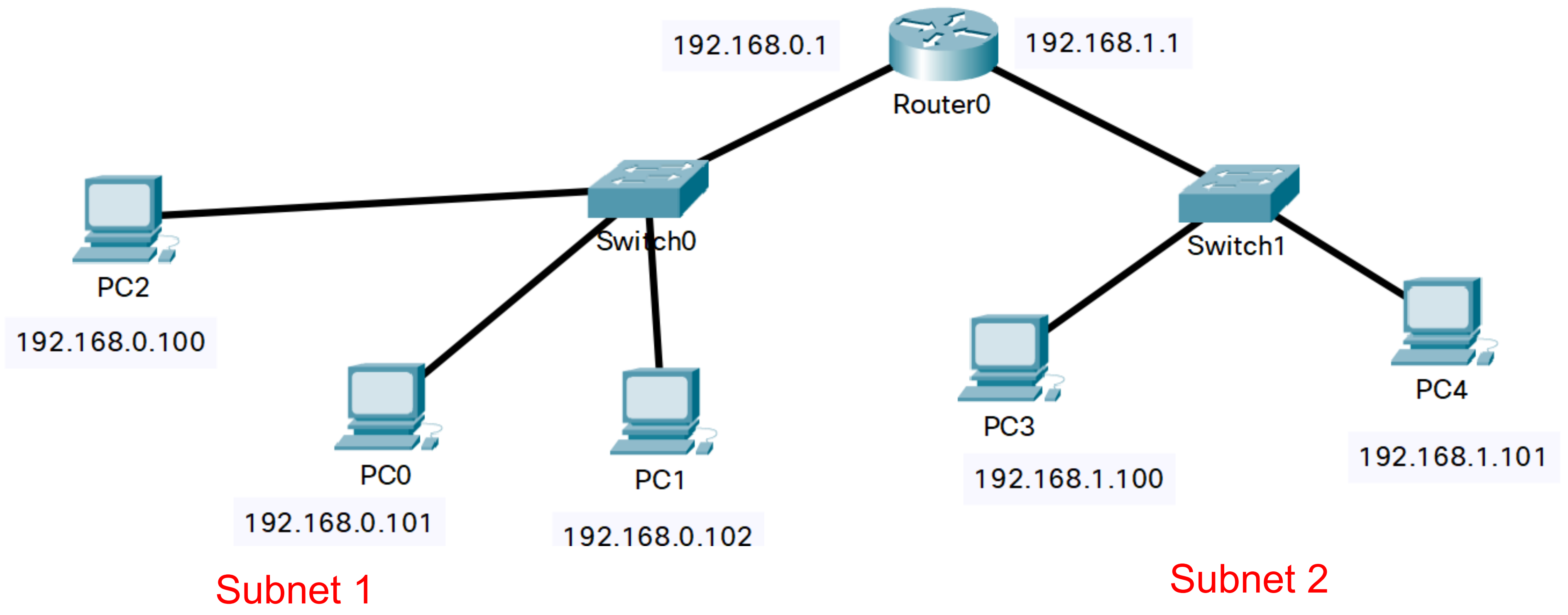
254 hosts

Subnet 2

192.168.1.0	Network address
192.168.1.1	First host address
192.168.1.254	Last host address
192.168.1.255	Subnet broadcast address

254 hosts

Assinging IP addresses to subnets



Reasons for Subnetting

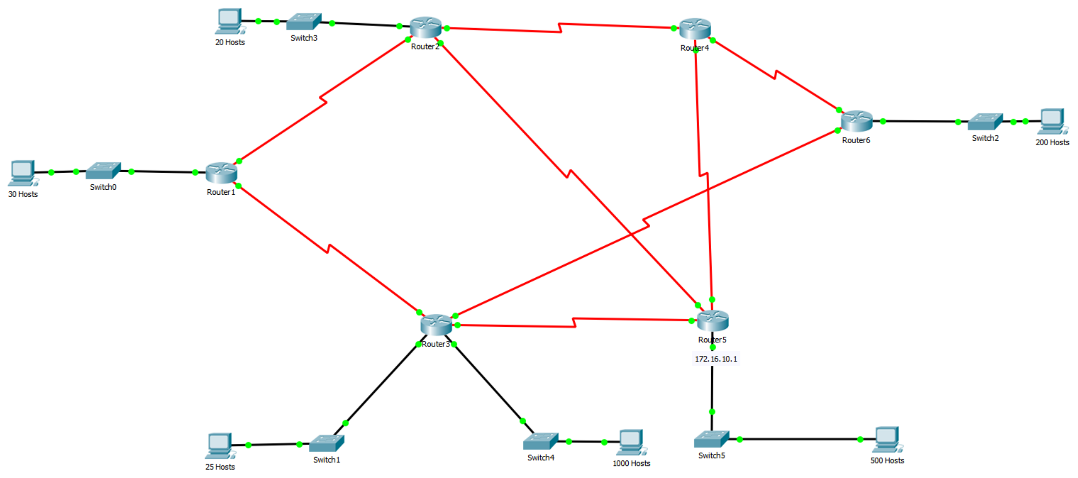
Large networks need to be segmented into smaller sub-networks, creating smaller groups of devices and services in order to:

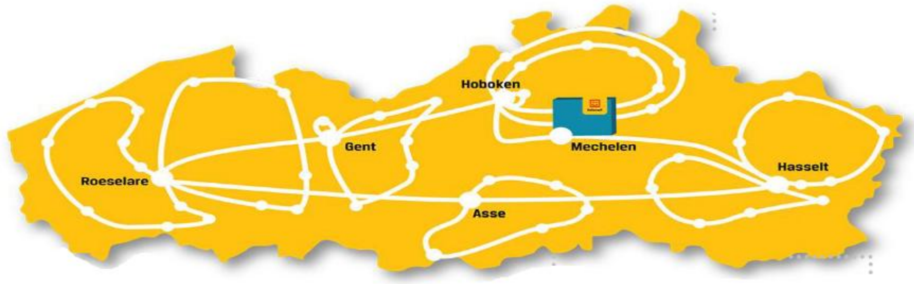
- Control traffic by containing broadcast traffic within subnetwork
- Reduce overall network traffic and improve network performance
- Security

Subnetting - process of segmenting a network into multiple smaller network spaces called subnetworks or **Subnets.**

Reasons for Subnetting

- Divide large networks into smaller networks
- Get IP address ranges for each network from a given address range

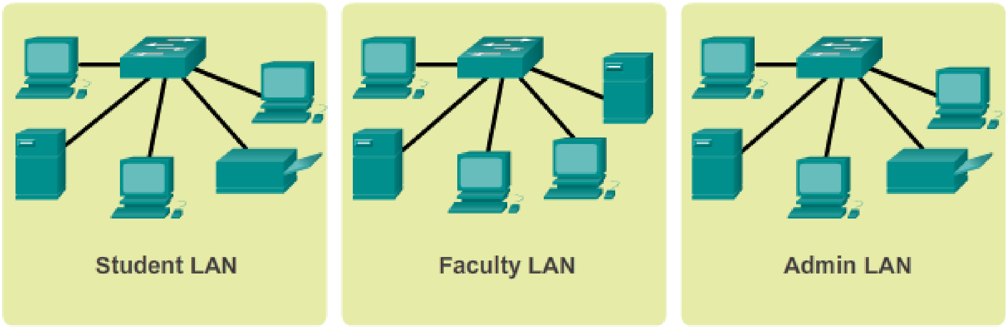




IP Subnetting is FUNdamental

The process of segmenting a network, by dividing it into multiple smaller network spaces, is called **subnetting**. These sub-networks are called **subnets**.
Network administrators can group devices and services into subnets that are determined by

- geographic location
- organizational unit (R&D,sales department,..)
- device type (printers, servers, WAN)
- any other division that makes sense for the network



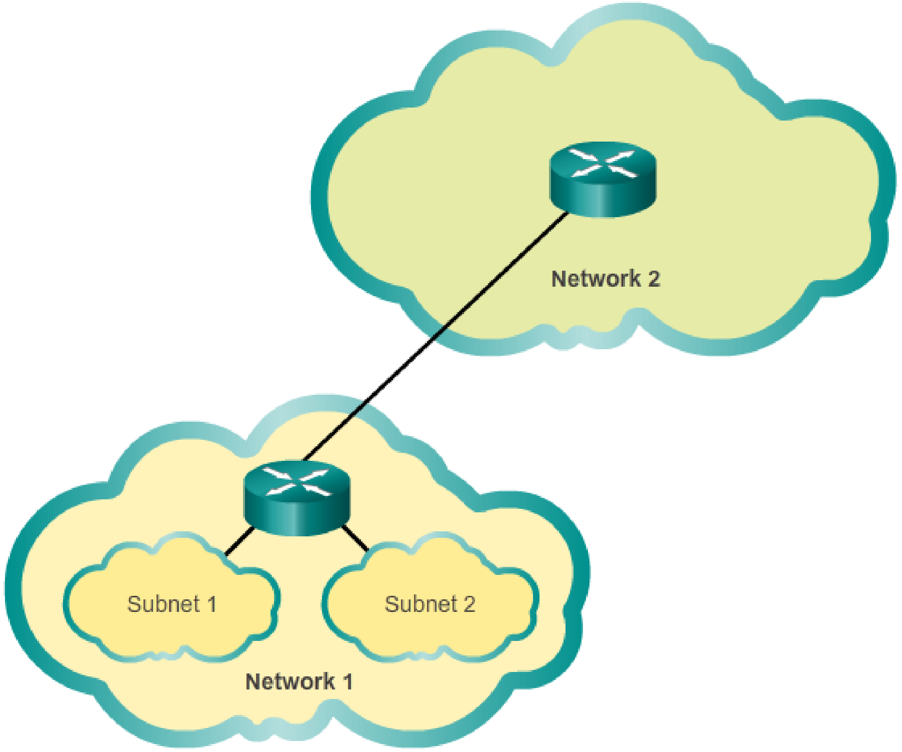
Planning requires decisions on each subnet in terms of size, the number of hosts per subnet, and how host addresses will be assigned.

IP Subnetting is <u>FUN</u>damental

Communication Between Subnets

- A router is necessary for devices on different networks and subnets to communicate.
- Each router interface must have an IPv4 host address that belongs to the network or subnet that the router interface is connected to.
- Devices on a network and subnet use the router interface attached to their LAN as their default gateway.

Communicating between Networks



Basic Subnetting

Remember the **private** IP address ranges are:

- 10.0.0.0 with a subnet mask of 255.0.0.0
- 172.16.0.0 with a subnet mask of 255.240.0.0
- 192.168.0.0 with a subnet mask of 255.255.0.0

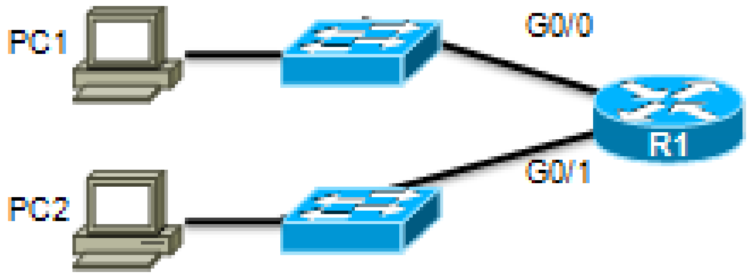
Create standards for IP address assignments within each subnet range. For example:

  - Printers and servers will be assigned static IP addresses
  - User will receive IP addresses from DHCP servers using /24 subnets
  - Routers are assigned the first available host addresses in the range

Basic Subnetting

Address	192	168	1	0000 0000
Mask	255	255	255	0000 0000

Network Portion Host Portion



Original	192.	168.	1.	0	000 0000
Mask	255.	255.	255.	0	000 0000

Network 192.168.1.0/24

Mask: 255.255.255.0

Borrowing 1 Bit from the host portion creates 2 subnets with the same subnet mask

Subnet 0

Network 192.168.1.**0-127**/**25**

Mask: 255.255.255.**128**

Subnet 1

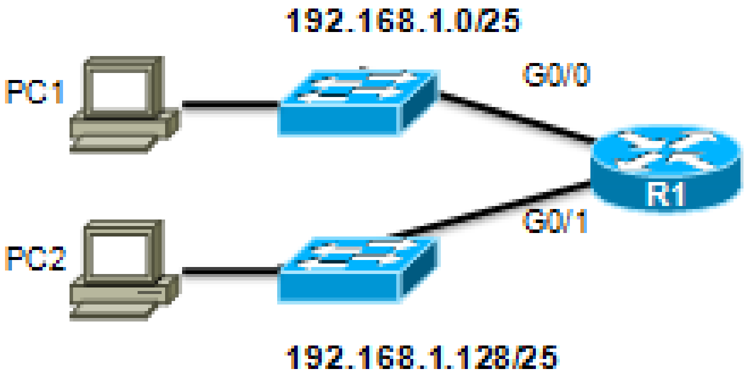
Network 192.168.1.**128-255**/**25**

Mask: 255.255.255.**128**

Subnets in Use

Subnet 0

Network 192.168.1.**0-127**/**25**



Subnet 1

Network 192.168.1.**128-255**/**25**

Address Range for 192.168.1.0/25 Subnet

Network Address	192.	168.	1.	0	000 0000	= 192.168.1.0
First Host Address	192.	168.	1.	0	000 0001	= 192.168.1.1
Last Host Address	192.	168.	1.	0	111 1110	= 192.168.1.126
Broadcast Address	192.	168.	1.	0	111 1111	= 192.168.1.127

Address Range for 192.168.1.128/25 Subnet

Network Address	192.	168.	1.	1	000 0000	= 192.168.1.128
First Host Address	192.	168.	1.	1	000 0001	= 192.168.1.129
Last Host Address	192.	168.	1.	1	111 1110	= 192.168.1.254
Broadcast Address	192.	168.	1.	1	111 1111	= 192.168.1.255

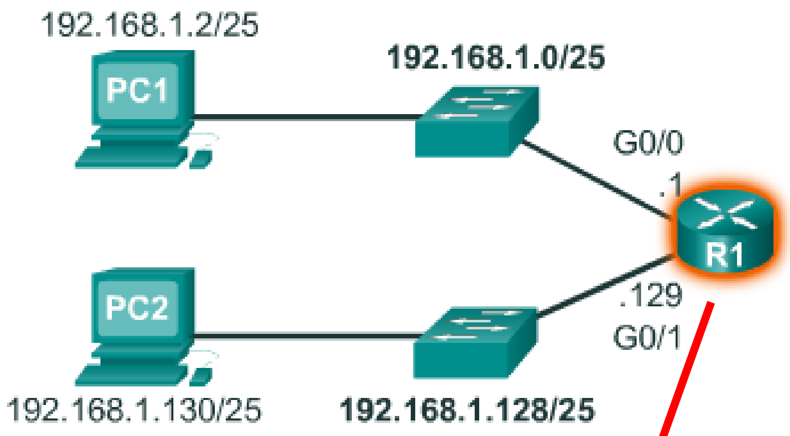
Subnets in Use

Subnet 0

Network 192.168.1.0-127/**25**

Subnet 1

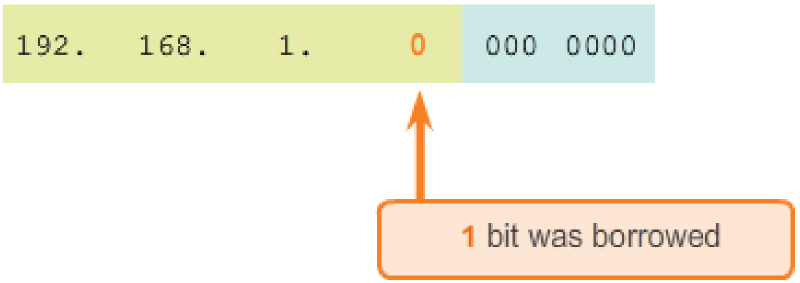
Network 192.168.1.128-255/**25**




```
R1(config)#interface gigabitethernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.128
R1(config-if)#exit
R1(config)#interface gigabitethernet 0/1
R1(config-if)#ip address 192.168.1.129 255.255.255.128
```

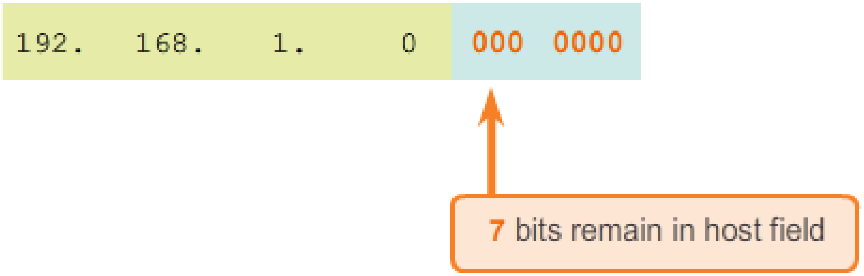
Calculate Number of Hosts

Subnets = 2^n
(where n = bits borrowed)



2^1 = 2 subnets

Hosts = 2^n
(where n = host bits remaining)



2^7 = 128 hosts per subnet
2^7 - 2 = 126 valid hosts per subnet

Creating 4 Subnets

Borrowing 2 Bits

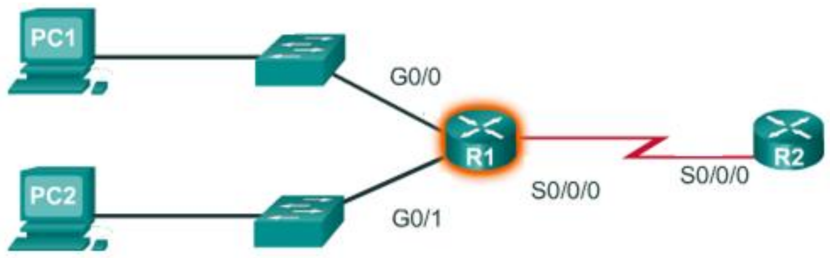
Original	192.	168.	1.	00	00	0000
Mask	255.	255.	255.	00	00	0000

Borrowing 2 bits creates 4 subnets:

Net 0	192.	168.	1.	00	00	0000	192.168.1.0/26
Net 1	192.	168.	1.	01	00	0000	192.168.1.64/26
Net 2	192.	168.	1.	10	00	0000	192.168.1.128/26
Net 3	192.	168.	1.	11	00	0000	192.168.1.192/26

All 4 subnets use the same mask:

Mask	255.	255.	255.	11	00	0000	Mask:255.255.255.192



Creating 4 Subnets

Address range for 192.168.0.1/26 subnet 00

Network Address	192.	168.	1.	00	00	0000	= 192.168.1.0	Network address
First Host Address	192.	168.	1.	00	00	0001	= 192.168.1.1	Usable hosts
Last Host Address	192.	168.	1.	00	11	1110	= 192.168.1.62	Usable hosts
Broadcast Address	192.	168.	1.	00	11	1111	= 192.168.1.63	Broadcast address

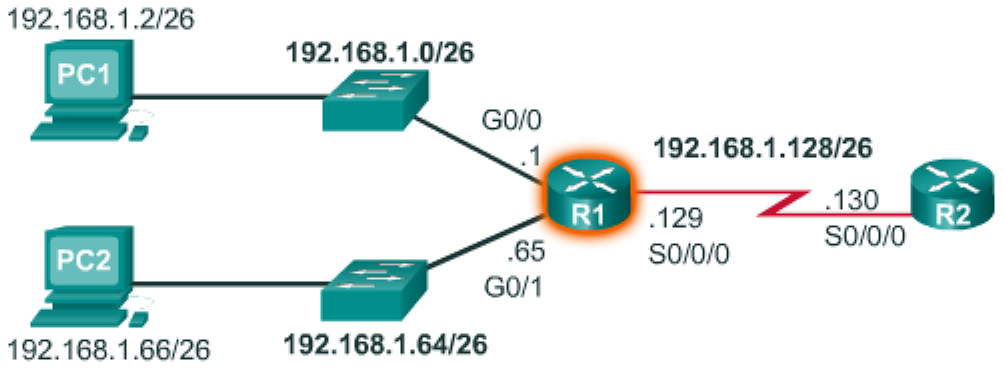
Creating 4 Subnets

Address Ranges Nets 0 - 2

Net 0	Network	192.	168.	1.	00	00	0000	192.168.1.0
	First	192.	168.	1.	00	00	0001	192.168.1.1
	Last	192.	168.	1.	00	11	1110	192.168.1.62
	Broadcast	192.	168.	1.	00	11	1111	192.168.1.63
Net 1	Network	192.	168.	1.	01	00	0000	192.168.1.64
	First	192.	168.	1.	01	00	0001	192.168.1.65
	Last	192.	168.	1.	01	11	1110	192.168.1.126
	Broadcast	192.	168.	1.	01	11	1111	192.168.1.127
Net 2	Network	192.	168.	1.	10	00	0000	192.168.1.128
	First	192.	168.	1.	10	00	0001	192.168.1.129
	Last	192.	168.	1.	10	11	1110	192.168.1.190
	Broadcast	192.	168.	1.	10	11	1111	192.168.1.191

Creating 4 Subnets

- 3 subnets configured on a router
- What is the 'problem' in this subnetting scenario?

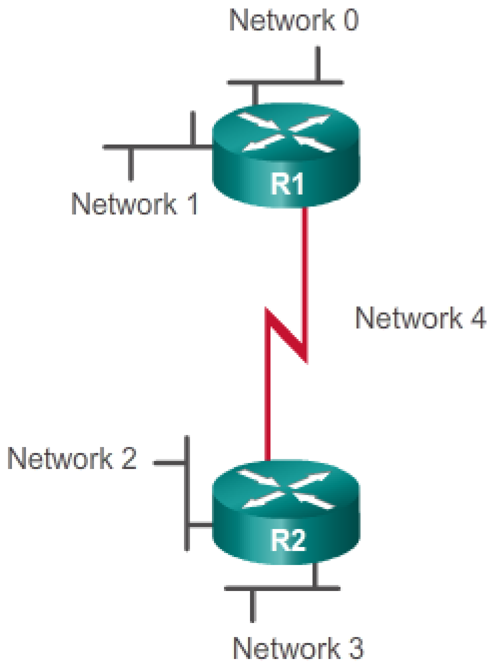



```
R1(config)#interface gigabitethernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.192
R1(config-if)#exit
R1(config)#interface gigabitethernet 0/1
R1(config-if)#ip address 192.168.1.65 255.255.255.192
R1(config-if)#exit
R1(config)#interface serial 0/0/0
R1(config-if)#ip address 192.168.1.129 255.255.255.192
```

Creating 8 Subnets

Borrowing 3 bits to Create
8 Subnets. 23 = 8 subnets

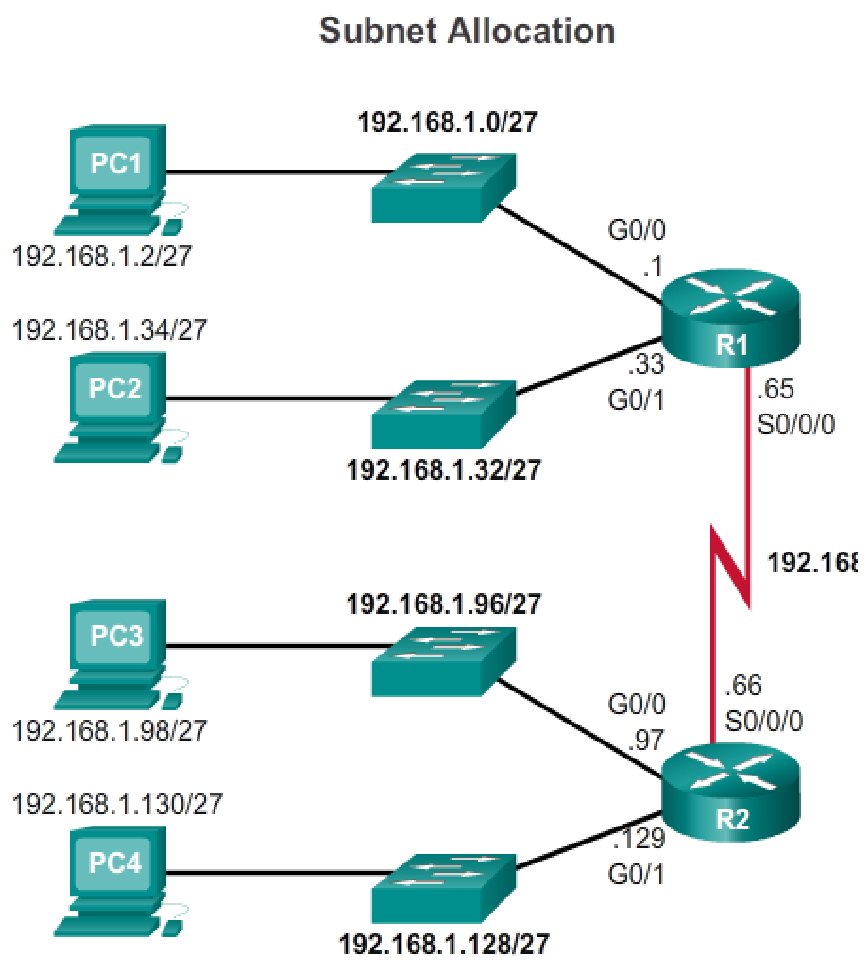
Net 0	Network	192.	168.	1.	000	0	0000	192.168.1.0
	First	192.	168.	1.	000	0	0001	192.168.1.1
	Last	192.	168.	1.	000	1	1110	192.168.1.30
	Broadcast	192.	168.	1.	000	1	1111	192.168.1.31
Net 1	Network	192.	168.	1.	001	0	0000	192.168.1.32
	First	192.	168.	1.	001	0	0001	192.168.1.33
	Last	192.	168.	1.	001	1	1110	192.168.1.62
	Broadcast	192.	168.	1.	001	1	1111	192.168.1.63
Net 2	Network	192.	168.	1.	010	0	0000	192.168.1.64
	First	192.	168.	1.	010	0	0001	192.168.1.65
	Last	192.	168.	1.	010	1	1110	192.168.1.94
	Broadcast	192.	168.	1.	010	1	1111	192.168.1.95
Net 3	Network	192.	168.	1.	011	0	0000	192.168.1.96
	First	192.	168.	1.	011	0	0001	192.168.1.97
	Last	192.	168.	1.	011	1	1110	192.168.1.126
	Broadcast	192.	168.	1.	011	1	1111	192.168.1.127



Creating 8 Subnets(continued)

Net 4	Network	192.	168.	1.	100	0	0000	192.168.1.128
	First	192.	168.	1.	100	0	0001	192.168.1.129
	Last	192.	168.	1.	100	1	1110	192.168.1.158
	Broadcast	192.	168.	1.	100	1	1111	192.168.1.159
Net 5	Network	192.	168.	1.	101	0	0000	192.168.1.160
	First	192.	168.	1.	101	0	0001	192.168.1.161
	Last	192.	168.	1.	101	1	1110	192.168.1.190
	Broadcast	192.	168.	1.	101	1	1111	192.168.1.191
Net 6	Network	192.	168.	1.	110	0	0000	192.168.1.192
	First	192.	168.	1.	110	0	0001	192.168.1.193
	Last	192.	168.	1.	110	1	1110	192.168.1.222
	Broadcast	192.	168.	1.	110	1	1111	192.168.1.223
Net 7	Network	192.	168.	1.	111	0	0000	192.168.1.224
	First	192.	168.	1.	111	0	0001	192.168.1.225
	Last	192.	168.	1.	111	1	1110	192.168.1.254
	Broadcast	192.	168.	1.	111	1	1111	192.168.1.255

Creating 8 Subnets(continued)



R1

```
R1(config)#interface gigabitethernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.224
R1(config-if)#exit
R1(config)#interface gigabitethernet 0/1
R1(config-if)#ip address 192.168.1.33 255.255.255.224
R1(config-if)#exit
R1(config)#interface serial 0/0/0
R1(config-if)#ip address 192.168.1.65 255.255.255.224
```

R2

```
R1(config)#interface gigabitethernet 0/0
R1(config-if)#ip address 192.168.1.97 255.255.255.224
R1(config-if)#exit
R1(config)#interface gigabitethernet 0/1
R1(config-if)#ip address 192.168.1.129 255.255.255.224
R1(config-if)#exit
R1(config)#interface serial 0/0/0
R1(config-if)#ip address 192.168.1.66 255.255.255.224
```

Subnetting

Host Address	192	168	60	26
Subnet Mask	255	255	255	240
Host Address in binary	11000000	10101000	00111100	00011010
Subnet Mask in binary	11111111	11111111	11111111	11110000
Network Address in binary				
Network Address in decimal				

Subnetting

Host Address	192	168	60	26
Subnet Mask	255	255	255	240
Host Address in binary	11000000	10101000	00111100	00011010
Subnet Mask in binary	11111111	11111111	11111111	11110000
Network Address in binary	**11000000**	**10101000**	**00111100**	**00010000**
Network Address in decimal	**192**	**168**	**60**	**16**

Network Address in decimal	192	168	238	12
Subnet Mask in decimal	255	255	255	252
Network address in binary	11000000	10101000	11101110	00001100
Subnet Mask in binary	11111111	11111111	11111111	11111100
First Usable Host IP Address in decimal				
Last Usable Host IP Address in decimal				
Broadcast Address in decimal				
Next Network Address in decimal				

Network Address in decimal	192	168	238	12
Subnet Mask in decimal	255	255	255	252
Network address in binary	11000000	10101000	11101110	00001100
Subnet Mask in binary	11111111	11111111	11111111	11111100
First Usable Host IP Address in decimal	**192**	**168**	**238**	**13**
Last Usable Host IP Address in decimal	**192**	**168**	**238**	**14**
Broadcast Address in decimal	**192**	**168**	**238**	**15**
Next Network Address in decimal	**192**	**168**	**238**	**16**

Network Address in decimal	192	168	52	0
Subnet Mask in decimal	255	255	255	192
Network address in binary	11000000	10101000	00110100	00000000
Subnet Mask in binary	11111111	11111111	11111111	11000000
First Usable Host IP Address in decimal				
Last Usable Host IP Address in decimal				
Broadcast Address in decimal				
Next Network Address in decimal				

Network Address in decimal	192	168	52	0
Subnet Mask in decimal	255	255	255	192
Network address in binary	11000000	10101000	00110100	00000000
Subnet Mask in binary	11111111	11111111	11111111	11000000
First Usable Host IP Address in decimal	**192**	**168**	**52**	**1**
Last Usable Host IP Address in decimal	**192**	**168**	**52**	**62**
Broadcast Address in decimal	**192**	**168**	**52**	**63**
Next Network Address in decimal	**192**	**168**	**52**	**64**

Subnetting Based on Host Requirements

There are two considerations when planning subnets:

Number of Subnets required

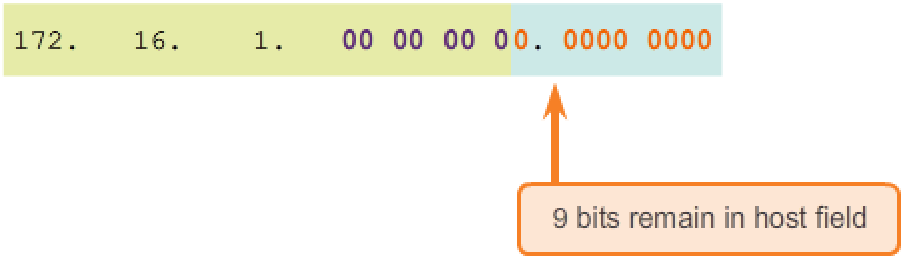
Number of Host addresses required

Formula to determine number of useable hosts

- **2^n-2**
- **2^n** (where n is the number the number of host bits remaining) is used to calculate the number of hosts
- **-2** Subnetwork address and broadcast address cannot be used on each subnet

Subnetting Based on Host Requirements

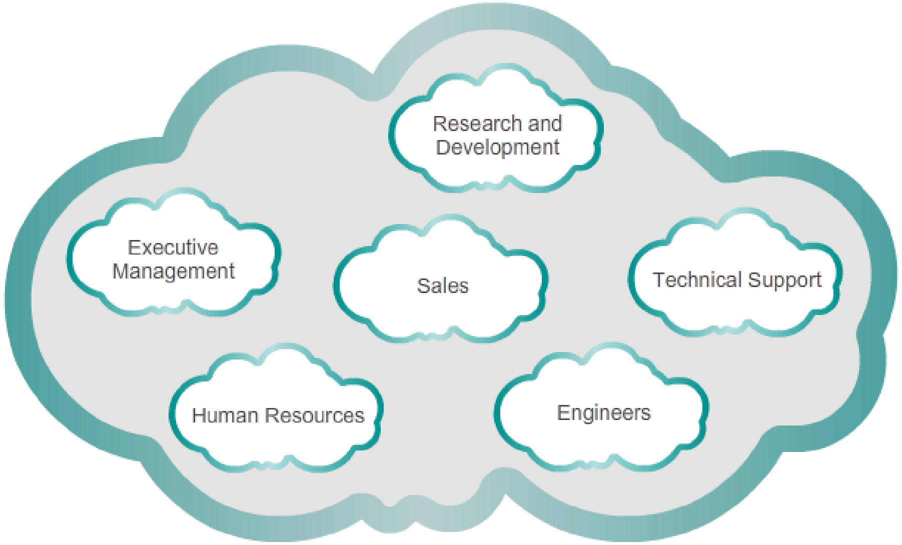
Hosts = 2^n
(where n = host bits remaining)



2^**9** = 512 hosts per subnet
2^**9** - 2 = 510 valid hosts per subnet

Subnetting Network-Based Requirements

Calculate number of subnets
Formula **2^n** (where n is the number of bits borrowed)
Subnet needed for each department in graphic



Subnetting Network-Based Requirements

172.	16.	0000 0000.	0000 0000	
255.	255.	1111 1110.	0000 0000	

Borrowing 7 bits creates 128 subnets

172.	16.	0000 0000.	0000 0000	172.16.0.0/23
172.	16.	0000 0010.	0000 0000	172.16.2.0/23
172.	16.	0000 0100.	0000 0000	172.16.4.0/23

. . to . .

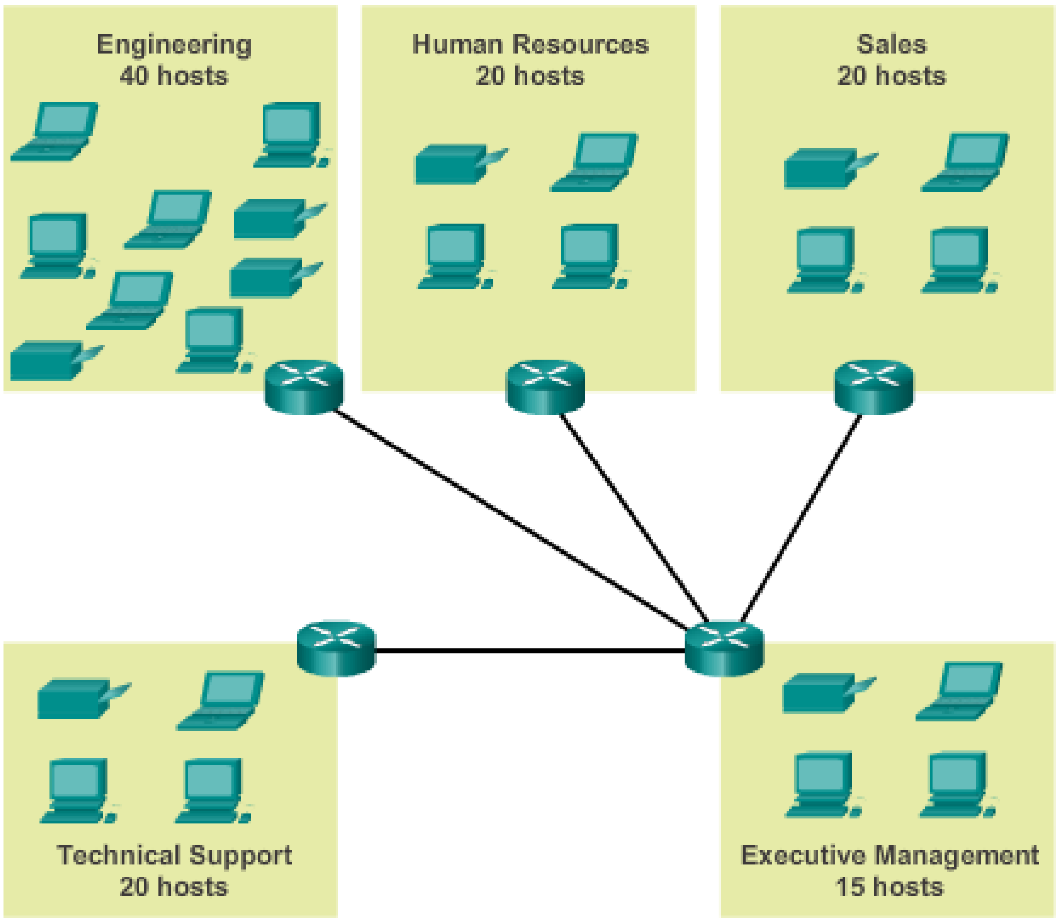
172.	16.	1111 1110.	0000 0000	172.16.254.0/23

Subnetting To Meet Network Requirements

- It is important to balance the number of subnets needed and the number of hosts required for the largest subnet.
- Design the addressing scheme to accommodate the maximum number of hosts for each subnet.
- Allow for growth in each subnet.
- Start with the largest subnet

Subnetting To Meet Network Requirements (cont)

- network topology consisting of 5 LAN segments and 4 internetwork connections between routers
- 9 subnets are required.
  This require 4 bit
  2^4 = 16
- The largest subnet requires 40 hosts.
  This require 6 bit
  2^6 -2 = 62



Subnetting To Meet Network Requirements (cont)

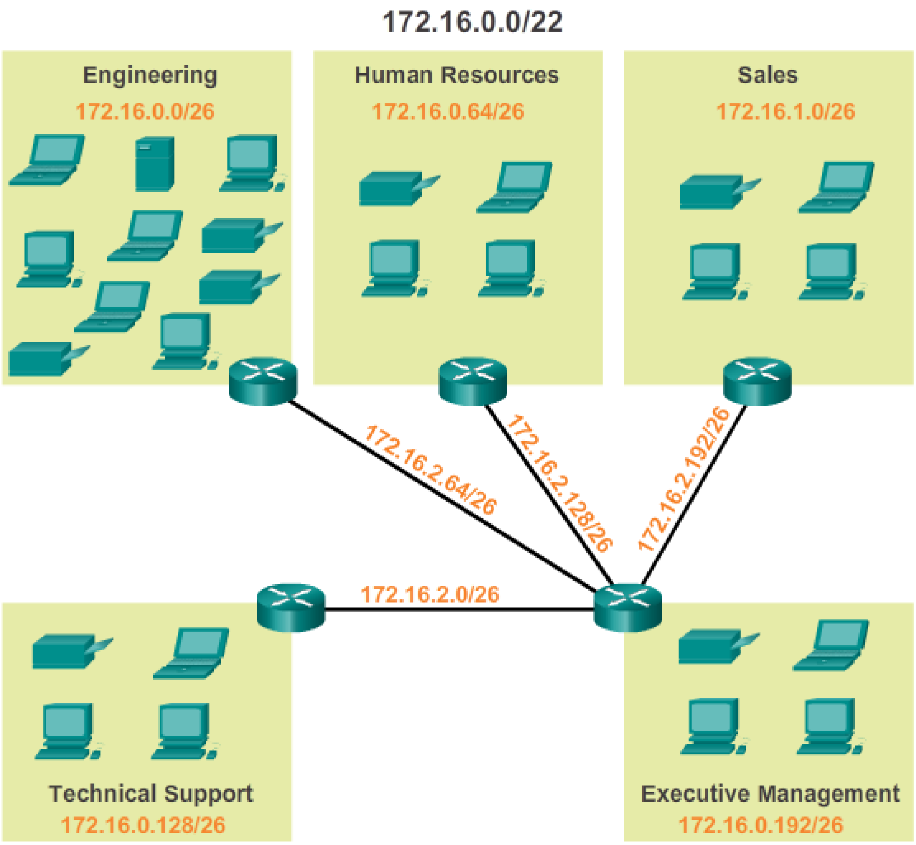
	10101100.00010000.00000000	00.00	000000	172.16.0.0/22
0	10101100.00010000.00000000	00.00	000000	172.16.0.0/26
1	10101100.00010000.00000000	00.01	000000	172.16.0.64/26
2	10101100.00010000.00000000	00.10	000000	172.16.0.128/26
3	10101100.00010000.00000000	00.11	000000	172.16.0.192/26
4	10101100.00010000.00000000	01.00	000000	172.16.1.0/26
5	10101100.00010000.00000000	01.01	000000	172.16.1.64/26
6	10101100.00010000.00000000	01.10	000000	172.16.1.128/26

Nets 7 – 14 not shown

15	10101100.00010000.00000000	11.10	000000	172.16.3.128/26
16	10101100.00010000.00000000	11.11	000000	172.16.3.192/26

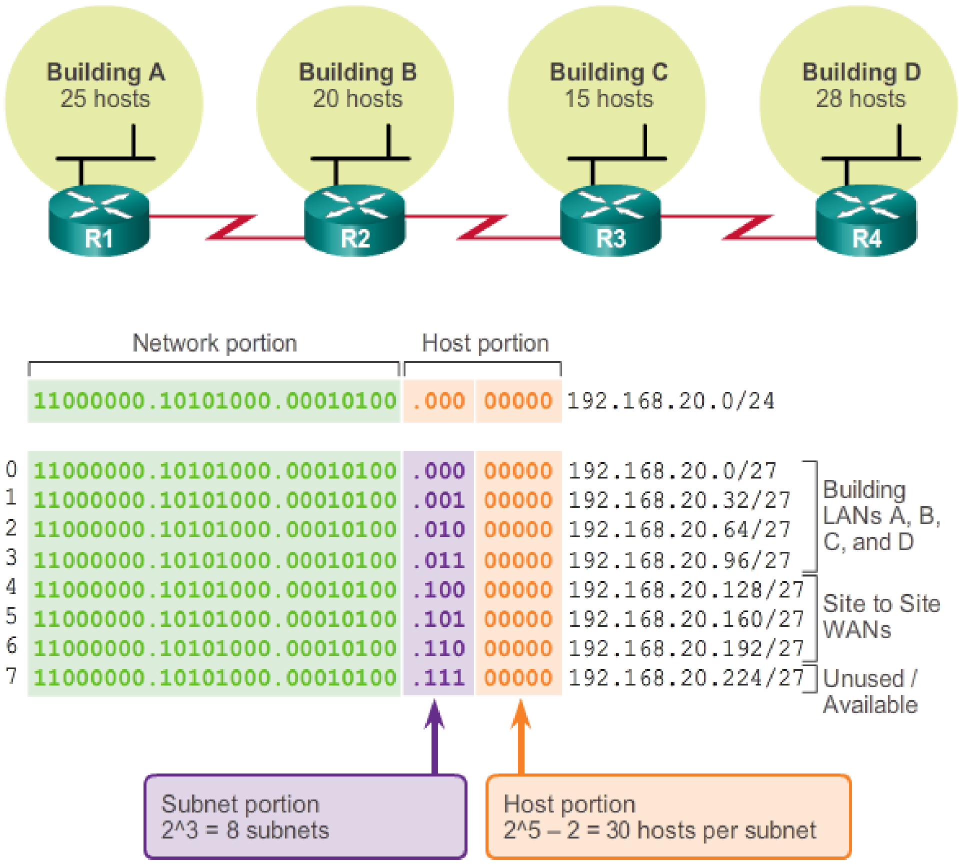
2^4 = 16 subnets

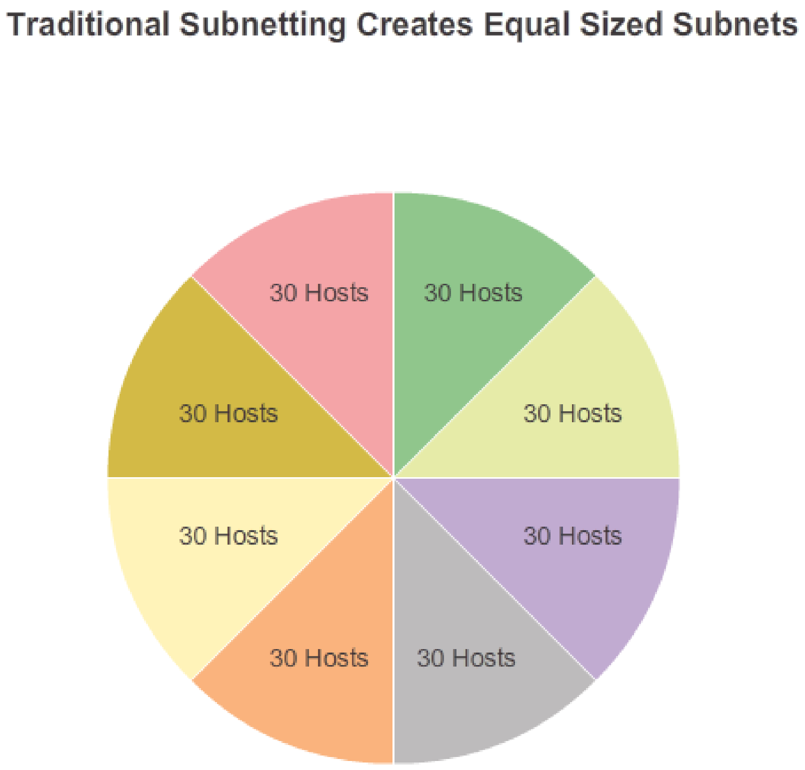
2^6 – 2 = 62 hosts per subnet



Waste of ip addresses

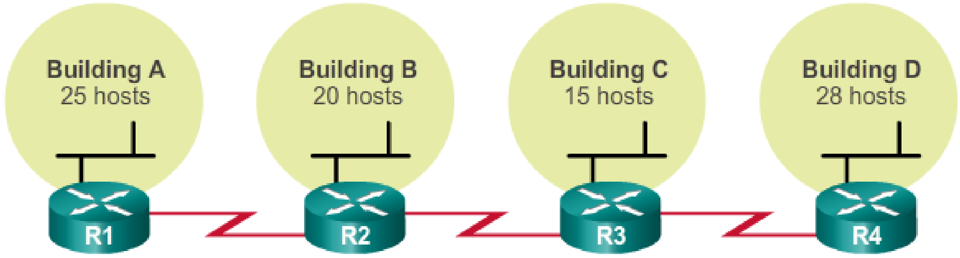
Subnetting with a fixed subnet mask for the entire network is wasteful
A lot of valid ip addresses are unused





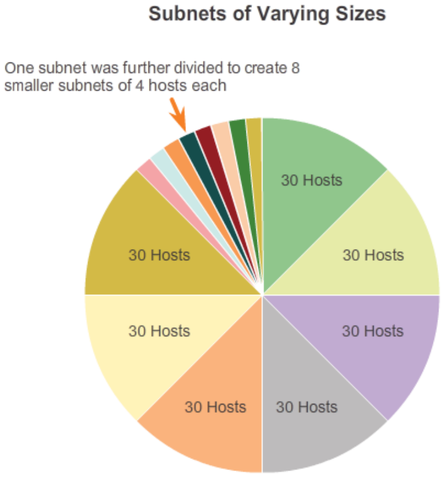
Variable Length Subnet Masks (VLSM)

VLSM allows a network space to be divided in **unequal parts**.
Subnet mask will vary depending on how many bits have been borrowed for a particular subnet.



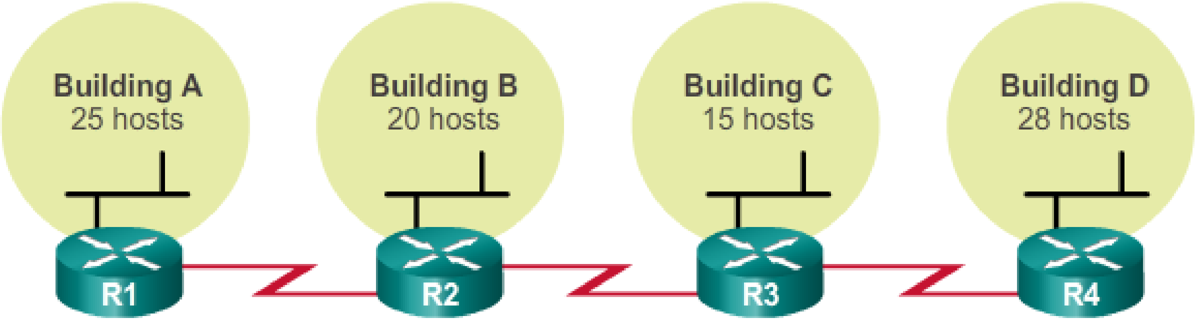

```
7   11000000.10101000.00010100.11100000  192.168.20.224/27

    3 more bits borrowed from subnet 7:

7:0 11000000.10101000.00010100.11100000  192.168.20.224/30
7:1 11000000.10101000.00010100.11100100  192.168.20.228/30
7:2 11000000.10101000.00010100.11101000  192.168.20.232/30
7:3 11000000.10101000.00010100.11101100  192.168.20.236/30
7:4 11000000.10101000.00010100.11110000  192.168.20.240/30
7:5 11000000.10101000.00010100.11110100  192.168.20.244/30
7:6 11000000.10101000.00010100.11111000  192.168.20.248/30
7:7 11000000.10101000.00010100.11111100  192.168.20.252/30
```

7:0 – 7:2	WANs
7:3 – 7:7	Unused / Available



How to VLSM

- Order your networks (subnets) from largest to smallest. Start with the largest network.
- Subnet the network for the largest subnet
- Go to the next network (subnet), see if you need to subnet again
- Repeat the process to create subnets of various sizes



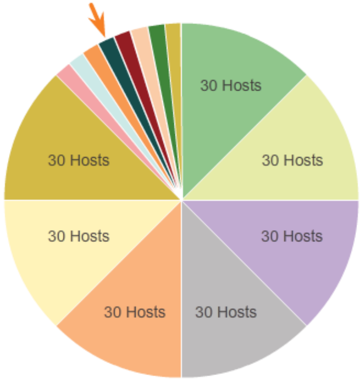
VLSM Subnetting of 192.168.20.0/24

	/27 Network	Hosts
Bldg A	.0	.1 - .30
Bldg B	.32	.33 - .62
Bldg C	.64	.65 - .94
Bldg D	.96	.97 - .126
Unused	.128	.129 - .158
Unused	.160	.161 - .190
Unused	.192	.193 - .222
	.224	.225 - .254

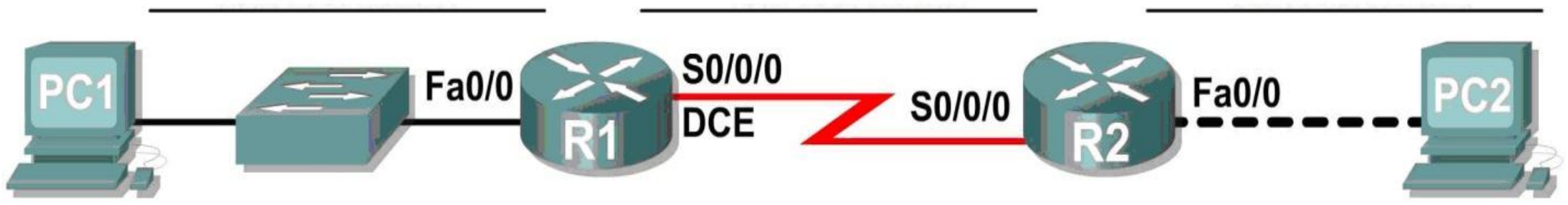
	/30 Network	Hosts
WAN R1–R2	.224	.225 - .226
WAN R2–R3	.228	.229 - .230
WAN R3–R4	.232	.233 - .234
Unused	.236	.237 - .238
Unused	.240	.241 - .242
Unused	.244	.245 - .246
Unused	.248	.249 - .250
Unused	.252	.253 - .254

Subnets of Varying Sizes

One subnet was further divided to create 8 smaller subnets of 4 hosts each



VLSM Example



Step 1: Examine the network requirements.
You have been given the 192.168.1.0/24 address space to use in your network design. The network consists of the following segments:

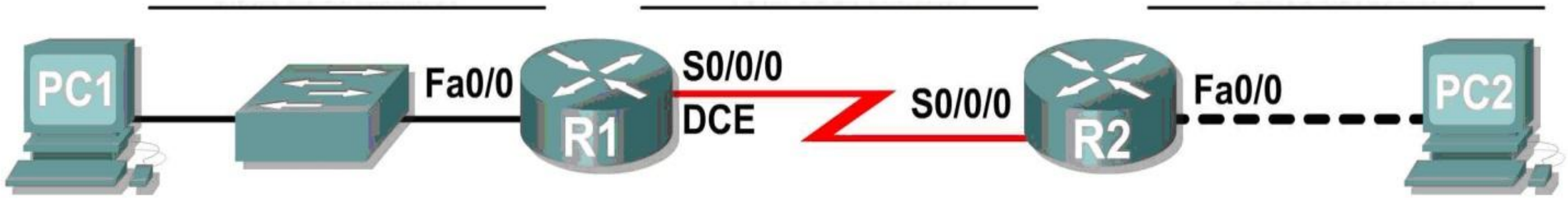
- The LAN connected to router R1 will require enough IP addresses to support **15 hosts**.
- The LAN connected to router R2 will require enough IP addresses to support **30 hosts.**
- The link between router R1 and router R2 will require IP addresses at each end of the link.

VLSM Example

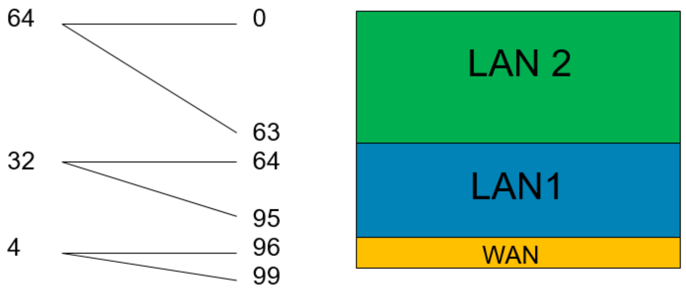
Usable address: 192.168.1.0/24

LAN2 = 30 hosts + router interface= 31 hosts = 6bit (2^6 – 2 = 62)				
Mask	255.255.255. (11000000)	255.255.255.192	/26	= 62 hosts
Network Address	192.168.1.(00000000)	192.168.1.0		
Broadcast Address	192.168.1.(00111111)	192.168.1.63		
Available Hosts	192.168.1.(00000001) – 192.168.1.(00111110)		192.168.1.1 – 192.168.1.62	
LAN1 = 15 hosts + router interface= 16 hosts = 5bit (2^5 – 2 = 30)				
Mask	255.255.255. (11100000)	255.255.255.224	/27	= 30 hosts
Network Address	192.168.1.(01000000)	192.168.1.64		
Broadcast Address	192.168.1.(01011111)	192.168.1.95		
Available Hosts	192.168.1.(01000001) – 192.168.1.(01011110)		192.168.1.65 – 192.168.1.94	
WAN = 2 hosts + router interface= 31 hosts = 2bit (2^2 – 2 = 2)				
Mask	255.255.255. (11111100)	255.255.255.252	/30	= 2 hosts
Network Address	192.168.1.(01100000)	192.168.1.96		
Broadcast Address	192.168.1.(01100011)	192.168.1.99		
Available Hosts	192.168.1.(01100001) – 192.168.1.(01100010)		192.168.1.97 – 192.168.1.98	

VLSM Example



Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.1.65	255.255.255.224	N/A
	S0/0/0	192.168.1.97	255.255.255.252	N/A
R2	Fa0/0	192.168.1.1	255.255.255.192	N/A
	S0/0/0	192.168.1.98	255.255.255.252	N/A
PC1	NIC	192.168.1.66	255.255.255.224	192.168.1.65
PC2	NIC	192.168.1.2	255.255.255.192	192.168.1.1



Planning to address a network

- Allocation of network addresses should be planned and documented for the purposes of:
- Preventing duplication of addresses
- Providing and controlling access
- Monitoring security and performance
- Addresses for Clients - usually dynamically assigned using Dynamic Host Configuration Protocol (DHCP)

Network: 192.168.1.0/24		
Use	**First**	**Last**
Host Devices	.1	.229
Servers	.230	.239
Printers	.240	.249
Intermediary Devices	.250	.253
Gateway (router LAN interface)	.254	